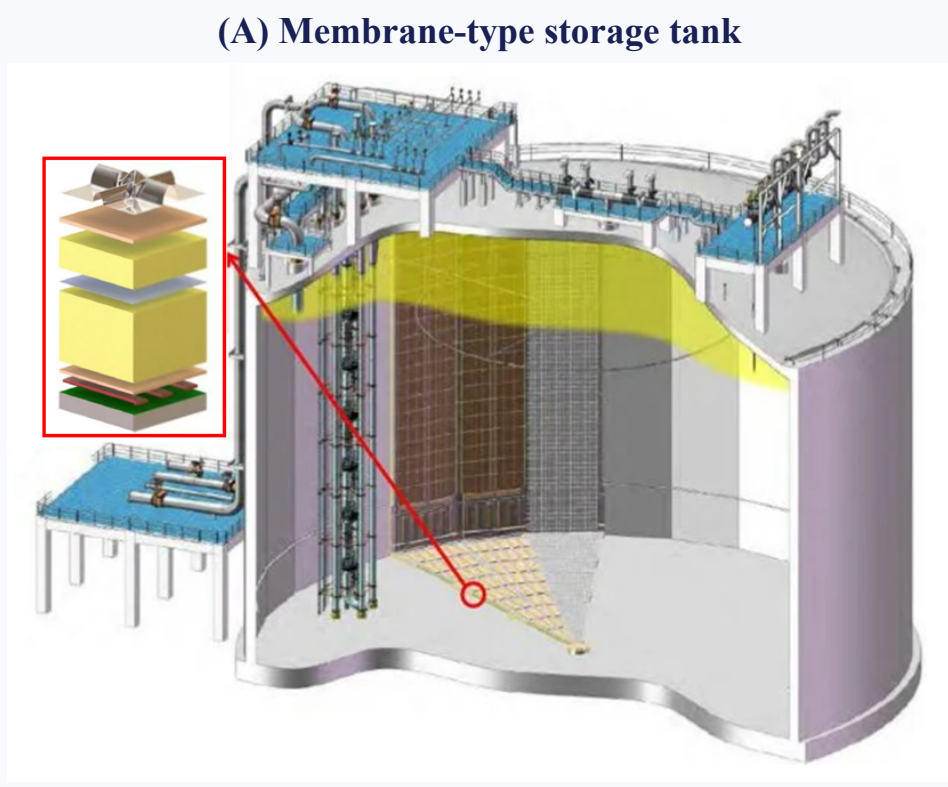




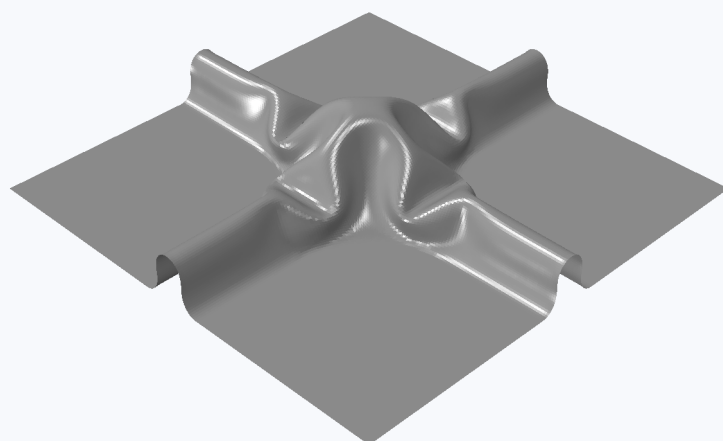
Introduction

Large-scale liquid hydrogen (LH₂) transport requires thin metallic containment barriers that remain gas-tight while accommodating thermal contraction and restrained in-plane deformation. Corrugated 316L stainless-steel membranes offer a promising solution because their geometry can redistribute imposed strain and reduce local deformation demand in nominally planar regions. Although the target application is cryogenic, room-temperature mechanical testing is used here to isolate the geometry-driven deformation-accommodation mechanism under controlled displacement loading, rather than to reproduce the full LH₂ service environment. Cryogenic integrity is examined separately through liquid-nitrogen (LN₂) cycling of a 0.2 m³ integrated demonstrator, while full mechanical qualification at LH₂ temperature remains a subsequent validation step.

This study experimentally evaluates the corrugated membrane concept using full-field digital image correlation (DIC) under tensile loading, acoustic-emission-monitored cyclic testing, and LN₂ thermal cycling with pressure-retention and leak-check assessment. The results clarify how corrugation features accommodate deformation and provide preliminary evidence for assembly-level integrity, supporting further development of corrugated membrane systems for large-scale LH₂ cargo containment.



(B) Corrugated membrane geometry



(C) Experimental validation route

	Purpose
Room-temperature DIC tensile test Full-field strain	Quantify corrugation-driven strain redistribution
AE-monitored cyclic loading Repeated mechanical loading	Assess cyclic response and activity evolution
LN₂ filling–draining cycles 0.2 m ³ integrated demonstrator	Evaluate response to cryogenic thermal cycling
Pressure hold + helium leak check Post-cycle integrity assessment	Verify pressure retention and gas-tightness